

Mangalayatan University Online
Assignment Cover Page- Jan'26 Session

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Course Name: Real Analysis
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Course Code: MMM-6114
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Date of Submission:

Note- The assignment Question have 5 Sections/Blocks. Kindly attempt Any **ONE question** from Each of the 5 blocks. Each question carries equal marks. For Eg:

Block 1 Attempt- Question 1a or 1b

Block 2 Attempt- Question 2a or 2b

Block 3 Attempt- Question 3a or 3b

Block 4 Attempt- Question 4a or 4b

Block 5 Attempt- Question 5a or 5b

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Type your assignment here...

or

You may attach handwritten assignment also (in PDF format)

Proof that the Set of Rational Numbers is not Order Complete

Statement

We shall prove that the ordered field $\mathbb{Q}$ of rational numbers is not order complete.

Recall the definition:

Definition (Order Completeness). An ordered set S is said to be order complete if every nonempty subset of S that is bounded above possesses a least upper bound (supremum) in S .

Thus, to show that $\mathbb{Q}$ is not order complete, it is enough to find:

- a nonempty subset $E \subset \mathbb{Q}$,
- such that E is bounded above in $\mathbb{Q}$,
- but E has no least upper bound in $\mathbb{Q}$.

Construction of the Set

Consider the set

$$E = \{x \in \mathbb{Q} : x^2 < 2\}.$$

We now analyze this set carefully.

Step 1: E is Nonempty

We observe that

$$1 \in E,$$

because

$$1^2 = 1 < 2.$$

Hence,

$$E \neq \emptyset.$$

Step 2: E is Bounded Above in $\mathbb{Q}$

We claim that 2 is an upper bound of E .

Indeed, let $x \in E$. Then

$$x^2 < 2.$$

Suppose $x \geq 2$. Then

$$x^2 \geq 4,$$

which contradicts $x^2 < 2$.

Therefore,

$$x < 2$$

for every $x \in E$.

Hence 2 is an upper bound of E in $\mathbb{Q}$.

Step 3: Assume E has a Supremum in $\mathbb{Q}$

Assume, for contradiction, that E possesses a least upper bound in $\mathbb{Q}$.

Let

$$\alpha = \sup E,$$

where $\alpha \in \mathbb{Q}$.

Since $1 \in E$ and α is an upper bound of E , we have

$$\alpha \geq 1.$$

Hence,

$$\alpha + 2 > 0.$$

We shall derive a contradiction.

Step 4: Analyze the Possible Values of α^2

There are three possibilities:

$$\alpha^2 < 2, \quad \alpha^2 > 2, \quad \alpha^2 = 2.$$

To analyze the first two cases, define

$$q = \alpha - \frac{\alpha^2 - 2}{\alpha + 2}.$$

Since $\alpha \in \mathbb{Q}$, it follows that $q \in \mathbb{Q}$.

Simplifying,

$$q = \frac{\alpha(\alpha + 2) - (\alpha^2 - 2)}{\alpha + 2} = \frac{2\alpha + 2}{\alpha + 2}.$$

Now compute:

$$q^2 - 2 = \left(\frac{2\alpha + 2}{\alpha + 2} \right)^2 - 2.$$

Therefore,

$$q^2 - 2 = \frac{4\alpha^2 + 8\alpha + 4 - 2(\alpha^2 + 4\alpha + 4)}{(\alpha + 2)^2}.$$

Hence,

$$q^2 - 2 = \frac{2(\alpha^2 - 2)}{(\alpha + 2)^2}.$$

Case 1: $\alpha^2 < 2$

Assume

$$\alpha^2 < 2.$$

Then

$$\alpha^2 - 2 < 0.$$

Since $\alpha + 2 > 0$,

$$\frac{\alpha^2 - 2}{\alpha + 2} < 0.$$

Hence,

$$q = \alpha - \frac{\alpha^2 - 2}{\alpha + 2} > \alpha.$$

Also,

$$q^2 - 2 = \frac{2(\alpha^2 - 2)}{(\alpha + 2)^2} < 0,$$

because $(\alpha + 2)^2 > 0$.

Thus,

$$q^2 < 2,$$

which implies

$$q \in E.$$

But $q > \alpha$, contradicting the fact that α is an upper bound of E .

Therefore,

$$\alpha^2 < 2$$

is impossible.

Case 2: $\alpha^2 > 2$

Assume

$$\alpha^2 > 2.$$

Then

$$\alpha^2 - 2 > 0.$$

Since $\alpha + 2 > 0$,

$$\frac{\alpha^2 - 2}{\alpha + 2} > 0.$$

Hence,

$$q = \alpha - \frac{\alpha^2 - 2}{\alpha + 2} < \alpha.$$

Also,

$$q^2 - 2 = \frac{2(\alpha^2 - 2)}{(\alpha + 2)^2} > 0.$$

Thus,

$$q^2 > 2.$$

Now let $x \in E$. Then

$$x^2 < 2 < q^2.$$

Since $q > 0$, it follows that

$$x < q.$$

Hence q is an upper bound of E .

But

$$q < \alpha,$$

contradicting the fact that α is the least upper bound of E .

Therefore,

$$\alpha^2 > 2$$

is impossible.

Case 3: $\alpha^2 = 2$

The only remaining possibility is

$$\alpha^2 = 2.$$

Assume

$$\alpha = \frac{m}{n},$$

where $m, n \in \mathbb{Z}$ have no common factor and $n \neq 0$.

Then

$$\left(\frac{m}{n}\right)^2 = 2,$$

so

$$m^2 = 2n^2.$$

Thus m^2 is even, which implies that m is even.

Let

$$m = 2k$$

for some integer k .

Substituting,

$$(2k)^2 = 2n^2,$$

which gives

$$4k^2 = 2n^2,$$

and hence

$$2k^2 = n^2.$$

Therefore n^2 is even, implying that n is even.

Thus both m and n are even, contradicting the assumption that $\frac{m}{n}$ is in lowest terms.

Hence no rational number satisfies

$$\alpha^2 = 2.$$

Therefore,

$$\alpha \notin \mathbb{Q},$$

contradicting our assumption that $\alpha \in \mathbb{Q}$.

Thus this case is also impossible.

Conclusion

All three possibilities lead to contradictions.

Hence our assumption that E possesses a supremum in $\mathbb{Q}$ is false.

Therefore, the set

$$E = \{x \in \mathbb{Q} : x^2 < 2\}$$

is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$.

Consequently,

$$\mathbb{Q}$$

is not order complete.

Theorem 1 (Darboux's Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Assume that the one-sided derivatives $f'(a)$ and $f'(b)$ exist.*

If k is any real number strictly between $f'(a)$ and $f'(b)$, then there exists a point $c \in (a, b)$ such that

$$f'(c) = k.$$

Proof. Without loss of generality, assume that

$$f'(a) < f'(b).$$

Let k be a real number satisfying

$$f'(a) < k < f'(b). \tag{1}$$

Our objective is to prove that there exists a point $c \in (a, b)$ such that

$$f'(c) = k.$$

Step 1: Define an Auxiliary Function

Define a function $g : [a, b] \rightarrow \mathbb{R}$ by

$$g(x) = f(x) - kx.$$

Since f is continuous on $[a, b]$ and differentiable on (a, b) , the same is true for g .

Differentiating,

$$g'(x) = f'(x) - k.$$

Step 2: Analyze the Endpoints

Evaluating at the endpoints:

1. At $x = a$,

$$g'(a) = f'(a) - k.$$

Using (1),

$$f'(a) < k,$$

hence

$$g'(a) < 0.$$

2. At $x = b$,

$$g'(b) = f'(b) - k.$$

Again using (1),

$$k < f'(b),$$

hence

$$g'(b) > 0.$$

Step 3: Apply the Extreme Value Theorem

Since g is continuous on the closed interval $[a, b]$, the Extreme Value Theorem guarantees that g attains an absolute minimum value on $[a, b]$.

Thus, there exists a point $c \in [a, b]$ such that

$$g(c) \leq g(x) \quad \text{for all } x \in [a, b].$$

Step 4: Show that the Minimum Cannot Occur at the End-points

We now prove that the minimum point c cannot be equal to a or b .

- **The minimum cannot occur at a .**

Since

$$g'(a) = \lim_{x \rightarrow a^+} \frac{g(x) - g(a)}{x - a} < 0,$$

there exists $\delta > 0$ such that whenever

$$a < x < a + \delta,$$

we have

$$\frac{g(x) - g(a)}{x - a} < 0.$$

Because $x - a > 0$, it follows that

$$g(x) - g(a) < 0,$$

and therefore

$$g(x) < g(a).$$

Hence $g(a)$ cannot be the absolute minimum value of g on $[a, b]$.

- **The minimum cannot occur at b .**

Similarly,

$$g'(b) = \lim_{x \rightarrow b^-} \frac{g(x) - g(b)}{x - b} > 0.$$

Therefore, there exists $\delta > 0$ such that whenever

$$b - \delta < x < b,$$

we have

$$\frac{g(x) - g(b)}{x - b} > 0.$$

Since $x - b < 0$, multiplying both sides by the negative quantity $(x - b)$ reverses the inequality:

$$g(x) - g(b) < 0.$$

Thus,

$$g(x) < g(b),$$

showing that $g(b)$ also cannot be the absolute minimum.

Since the minimum cannot occur at either endpoint, it must occur at some interior point

$$c \in (a, b).$$

Step 5: Apply Fermat's Theorem

Because g attains a local minimum at the interior point c and g is differentiable at c , Fermat's Theorem implies that

$$g'(c) = 0.$$

Substituting the definition of $g'(x)$,

$$f'(c) - k = 0.$$

Hence,

$$f'(c) = k.$$

This completes the proof.

□

Uniform Convergence of $f_n(x) = x^n$

Proof with Complete Justifications

Problem Statement

Let $\langle f_n \rangle$ be a sequence of functions defined by $f_n(x) = x^n$ for $n \in \mathbb{N}$. Show that:

1. $\langle f_n \rangle$ is uniformly convergent on $[0, k]$ where $0 < k < 1$.
2. $\langle f_n \rangle$ is only pointwise convergent (not uniformly convergent) on $[0, 1]$.

Proof

Step 1: Pointwise Limit Function

We first determine the pointwise limit of $f_n(x) = x^n$ on $[0, 1]$.

- For $x \in [0, 1)$: Since $0 \leq x < 1$, we have $\lim_{n \rightarrow \infty} x^n = 0$.
- For $x = 1$: $\lim_{n \rightarrow \infty} 1^n = 1$.

Thus the pointwise limit function $f : [0, 1] \rightarrow \mathbb{R}$ is:

$$f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

Step 2: Uniform Convergence on $[0, k]$ where $0 < k < 1$

We apply the **Supremum (M_n) Test**: a sequence $\langle f_n \rangle$ converges uniformly to f on a set S if and only if

$$M_n := \sup_{x \in S} |f_n(x) - f(x)| \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since $x \in [0, k]$ with $k < 1$, the pointwise limit is $f(x) = 0$ throughout $[0, k]$. Therefore:

$$M_n = \sup_{x \in [0, k]} |x^n - 0| = \sup_{x \in [0, k]} x^n.$$

Claim: $f_n(x) = x^n$ is monotonically increasing on $[0, k]$.

Justification. Differentiating, $f'_n(x) = nx^{n-1}$. For $x \in [0, k]$ with $k > 0$ and $n \geq 1$, we have $x \geq 0$, so $f'_n(x) = nx^{n-1} \geq 0$. Hence f_n is non-decreasing on $[0, k]$, and strictly increasing on $(0, k]$. $\square$

Since f_n is increasing on $[0, k]$, its supremum is attained at the right endpoint $x = k$:

$$M_n = k^n.$$

Taking the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} M_n = \lim_{n \rightarrow \infty} k^n = 0, \quad \text{since } 0 < k < 1.$$

By the M_n -test, $\langle f_n \rangle$ **converges uniformly** to $f(x) = 0$ on $[0, k]$. □

Step 3: Only Pointwise Convergence on $[0, 1]$

We now show that the convergence *fails* to be uniform on the full interval $[0, 1]$. We use the contrapositive of the following classical result.

Theorem 1 (Continuity of Uniform Limits). *Let $\langle f_n \rangle$ be a sequence of functions, each continuous on an interval $I \subseteq \mathbb{R}$. If $f_n \rightarrow f$ uniformly on I , then f is continuous on I .*

We apply the **contrapositive**: if the limit function f is *discontinuous* on I , then the convergence cannot be uniform on I .

Proof of Non-Uniformity on $[0, 1]$ Each $f_n(x) = x^n$ is a polynomial, hence continuous on $[0, 1]$ for every $n \in \mathbb{N}$.

However, the pointwise limit f established in Step 1 is *discontinuous* at $x = 1$, since:

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} 0 = 0 \neq 1 = f(1).$$

By the contrapositive of Theorem 1, a sequence of continuous functions cannot converge uniformly to a discontinuous limit. Therefore, $\langle f_n \rangle$ does **not** converge uniformly on $[0, 1]$.

Hence $\langle f_n \rangle$ is only **pointwise convergent** on $[0, 1]$. □

Remark. The non-uniform convergence can also be verified directly: choose $x_n = k^{1/n}$ for any fixed $0 < k < 1$, so $x_n \in [0, 1)$ and $x_n \rightarrow 1$. Then $f_n(x_n) = (k^{1/n})^n = k \not\rightarrow 0 = f(x_n)$, confirming that no single N works for all x near 1.

Analysis of $f(x, y) = \sqrt{|xy|}$ at the Origin

Problem Statement

Prove that the function,

$$f(x, y) = \sqrt{|xy|}.$$

is not differentiable at the point $(0,0)$, but that f_x and f_y both exist at the origin and have the value 0. Hence deduce that these two partial derivatives are continuous except at the origin.

Solution

Study the following at $(0,0)$:

1. Existence of the partial derivatives $f_x(0,0)$ and $f_y(0,0)$.
2. Differentiability of f at $(0,0)$.
3. Continuity of the partial derivatives.

Step 1: Partial Derivatives at $(0,0)$

Using the limit definition,

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{|h \cdot 0|}}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0.$$

By symmetry,

$$f_y(0,0) = 0.$$

Step 2: Differentiability at $(0,0)$

If f were differentiable at $(0,0)$, then since $f(0,0) = 0$ and both partial derivatives vanish there, the following limit would equal zero:

$$L = \lim_{(h,k) \rightarrow (0,0)} \frac{f(h,k) - f(0,0) - f_x(0,0)h - f_y(0,0)k}{\sqrt{h^2 + k^2}} = \lim_{(h,k) \rightarrow (0,0)} \frac{\sqrt{|hk|}}{\sqrt{h^2 + k^2}}.$$

Take the path $k = mh$ ($m \neq 0$, $h \rightarrow 0$):

$$\frac{\sqrt{|h \cdot mh|}}{\sqrt{h^2 + m^2h^2}} = \frac{|h|\sqrt{|m|}}{|h|\sqrt{1+m^2}} = \frac{\sqrt{|m|}}{\sqrt{1+m^2}}.$$

For $m = 1$ this gives $\frac{1}{\sqrt{2}} \neq 0$.

Since the limit depends on the path, $L \neq 0$, and therefore f is **not differentiable** at $(0,0)$.

Step 3: Partial Derivatives Away from the Axes

For $xy \neq 0$, the chain rule gives

$$f_x(x, y) = \frac{1}{2}|xy|^{-1/2} \cdot \frac{\partial}{\partial x}|xy| = \frac{y \operatorname{sgn}(xy)}{2\sqrt{|xy|}},$$

or equivalently,

$$f_x(x, y) = \frac{\operatorname{sgn}(x)\sqrt{|y|}}{2\sqrt{|x|}}.$$

Similarly,

$$f_y(x, y) = \frac{x \operatorname{sgn}(xy)}{2\sqrt{|xy|}} = \frac{\operatorname{sgn}(y)\sqrt{|x|}}{2\sqrt{|y|}}.$$

Step 4: Continuity of the Partial Derivatives

Non-continuity at $(0, 0)$. Consider f_x as $(x, y) \rightarrow (0, 0)$ along two paths.

- Along $y = x^2$ ($x > 0$):

$$f_x(x, x^2) = \frac{\sqrt{x^2}}{2\sqrt{x}} = \frac{x}{2\sqrt{x}} = \frac{\sqrt{x}}{2} \rightarrow 0.$$

- Along $y = x$ ($x > 0$):

$$f_x(x, x) = \frac{\sqrt{x}}{2\sqrt{x}} = \frac{1}{2}.$$

The two path limits differ, so $\lim_{(x,y) \rightarrow (0,0)} f_x(x, y)$ does not exist. Hence f_x is **not continuous at $(0, 0)$** .

Non-existence on the coordinate axes. Consider the point $(0, 1)$. By definition,

$$f_x(0, 1) = \lim_{h \rightarrow 0} \frac{f(h, 1) - f(0, 1)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{|h|}}{h}.$$

For $h > 0$ this equals $h^{-1/2} \rightarrow +\infty$, so the limit does not exist.

In general, $f_x(0, y)$ does not exist for any $y \neq 0$, and by symmetry, $f_y(x, 0)$ does not exist for any $x \neq 0$.

Summary.

Point	f_x exists?	f_y exists?
$xy \neq 0$	Yes	Yes
$(0, 0)$	Yes ($= 0$)	Yes ($= 0$)
$x = 0, y \neq 0$	No	Yes
$y = 0, x \neq 0$	Yes	No

Conclusion. The partial derivatives f_x and f_y :

- exist and are continuous on the open region $\{(x, y) : xy \neq 0\}$,
- exist (and equal zero) at $(0, 0)$, but are **not continuous** there,
- **do not exist** on most of the coordinate axes (excluding the origin).

Hence the claim that “the partial derivatives are continuous everywhere except at $(0, 0)$ ” is **incorrect**; they fail to exist on the entire y -axis (excluding the origin) for f_x , and on the entire x -axis (excluding the origin) for f_y .

Proof of an Integral Inequality

Problem Statement

Prove that:

$$\frac{\pi^2}{9} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{2\pi^2}{9}.$$

Proof

Key Lemma: Monotonicity of $x/\sin x$ on $(0, \pi)$

Lemma 1. *The function $g(x) = \frac{x}{\sin x}$ is strictly increasing on $(0, \pi)$.*

Proof. Differentiating,

$$g'(x) = \frac{\sin x - x \cos x}{\sin^2 x}.$$

It suffices to show the numerator is positive. Define $\phi(x) = \sin x - x \cos x$. Then $\phi(0) = 0$ and

$$\phi'(x) = \cos x - \cos x + x \sin x = x \sin x > 0 \quad \text{for all } x \in (0, \pi),$$

since both $x > 0$ and $\sin x > 0$ on that interval. Hence $\phi(x) > 0$ on $(0, \pi)$, so $g'(x) > 0$, and g is strictly increasing on $(0, \pi)$. $\square$

Step 1: Bounds on the integrand

Since $[\pi/6, \pi/2] \subset (0, \pi)$ and g is strictly increasing on $(0, \pi)$ by Lemma 1, its minimum and maximum on $[\pi/6, \pi/2]$ are attained at the left and right endpoints respectively:

$$g\left(\frac{\pi}{6}\right) = \frac{\pi/6}{\sin(\pi/6)} = \frac{\pi/6}{1/2} = \frac{\pi}{3},$$

$$g\left(\frac{\pi}{2}\right) = \frac{\pi/2}{\sin(\pi/2)} = \frac{\pi/2}{1} = \frac{\pi}{2}.$$

Therefore, for all $x \in [\pi/6, \pi/2]$:

$$\frac{\pi}{3} \leq \frac{x}{\sin x} \leq \frac{\pi}{2}.$$

Step 2: Apply the comparison theorem for integrals

Theorem (Integral comparison). If $m \leq f(x) \leq M$ for all $x \in [a, b]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a).$$

The length of the interval is

$$b-a = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi - \pi}{6} = \frac{\pi}{3}.$$

Applying the theorem with $m = \pi/3$, $M = \pi/2$, and $b-a = \pi/3$:

$$\frac{\pi}{3} \cdot \frac{\pi}{3} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{\pi}{2} \cdot \frac{\pi}{3}.$$

Step 3: Evaluate and compare the bounds

- Lower bound:

$$\frac{\pi}{3} \cdot \frac{\pi}{3} = \frac{\pi^2}{9}.$$

- Upper bound:

$$\frac{\pi}{2} \cdot \frac{\pi}{3} = \frac{\pi^2}{6}.$$

This gives us:

$$\frac{\pi^2}{9} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{\pi^2}{6}.$$

Since $\frac{\pi^2}{6} \leq \frac{2\pi^2}{9}$ (equivalent to $\frac{1}{6} \leq \frac{2}{9}$, i.e. $3 \leq 4$, which is true), the upper bound implies:

$$\int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{\pi^2}{6} \leq \frac{2\pi^2}{9}.$$

Conclusion

Combining the lower and upper bounds:

$$\boxed{\frac{\pi^2}{9} \leq \int_{\pi/6}^{\pi/2} \frac{x}{\sin x} dx \leq \frac{2\pi^2}{9}}.$$

□